

Abstract

Title: AI-Based Virtual Insurance Agent for Enhanced Insurance Accessibility in Rural Areas

Insurance plays a vital role in providing financial security against health, agricultural, and life-related risks. However, in rural India, insurance awareness and accessibility remain low due to complex policy terms, limited access to advisors, and language and literacy barriers. To address this challenge, this mini project is an AI-powered virtual insurance agent designed to enhance insurance accessibility for rural communities through a digital platform. The system uses a conversational interface that allows users to interact via text or voice to obtain clear and simplified explanations of insurance concepts. A Retrieval-Augmented Generation (RAG) approach is employed to ensure accurate and reliable responses by grounding the AI's output in verified insurance documents and government scheme information. Additionally, a machine learning-based recommendation module analyses basic user profile data to suggest suitable insurance categories tailored to individual needs. The proposed solution integrates a lightweight frontend interface with a FastAPI-based backend and an AI layer composed of a pre-trained language model, document retrieval mechanisms, and an explainable recommendation engine. By combining accessibility, accuracy, and personalization, the system aims to bridge the information gap in rural insurance adoption and demonstrate the effective use of AI for social impact. The project highlights a scalable and practical approach to deploying intelligent digital assistants in real-world public welfare domains.

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